

UKRAINE'S USV: The maritime drones used to attack the Russian fleet in Sevastopol in October 2022 were each 5.5 m long, with a range of 40 km, 60-hr endurance and warhead load of up to 200 kilos



US NAVY'S T-38 DEVIL RAY USV Developed by MARTAC, the T-38 Devil Ray has a speed of 25 knots and can carry a payload of 2,050 kg



CHINA'S SEA WING UNDERWATER GLIDERS These are a type of UUVs that are unpowered and employ the variable-buoyancy propulsion system. They also have large wings that enable them to glide forward over vast distances

ACCORDING TO TOP NAVAL OFFICIALS, AUTONOMOUS AND UNMANNED SYSTEMS WILL FUNDAMENTALLY CHANGE THE WAY NAVAL WARFARE IS CONDUCTED

remotely and navigate autonomously. The 88.5m-long craft can carry drones, UUVs and USVs on board.

DOMESTIC PUSH

While the Integrated Unmanned Roadmap of the Indian Navy of 2021 was a comprehensive charter in consonance with the Indian Navy's 'Concept of Operations', the navy's Vision Document for Unmanned Systems 2022-30 is a capability development document to guide stakeholders who possess the resources to accelerate the development of unmanned systems.

Likewise, multiple Indian start-ups and firms in the MSME sector, along with the Defence Research and Development Organisation (DRDO), are working on developing indigenous technology for unmanned platforms. Currently, the navy is steering five projects involving private industry, all aiming to bolster *atmanirbharta* (self-sufficiency) in the unmanned domain—unmanned high endurance autonomous underwater vessels (HEAUVs), autonomous surface vessels, semi-submersible autonomous vessels and buoyancy underwater gliders (a type of UUV). Two such firms in an advanced stage of completing their projects are Tardid Technology and Sagar Defence Engineering Ltd. The navy aims to induct all these platforms within the next few years.

Aastha Verma, COO of Bengaluru-based Tardid Technology, believes that unmanned surface vessels are emerging as the future of maritime security in the Indian Ocean. "These unmanned vessels, equipped with state-of-the-art technology, can operate in remote and harsh environments, providing real-time intelligence, surveillance and reconnaissance capabilities," she says. The USV Tardid is developing, Brainbox Smart Ship Operations, has a length of about eight metres, an endurance (the maximum time it can operate without being refuelled) of over 12 hours and can travel at a speed of 12 nautical miles per hour. Brainbox units being developed for the navy are for reconnaissance, though they can be weaponised, says Verma.

Pune-based Sagar Defence Engineering Ltd (SDEL) is working with the navy to develop India's first armed autonomous uncrewed boat with swarming capabilities. Swarming technology involves several 'networked' or interconnected vessels that perform a task in clusters—be it minesweeping, intelligence-gathering, surveillance or striking enemy targets. The Indian Navy is using the USV project to better understand and evaluate the capabilities of USV swarms. If successful, high numbers can be ordered.

Captain Nikunj Parashar of SDEL says that full autonomous capability involves AI-based Collision Avoidance (CA) and smart mission capability suites. The USV, he says, includes an onboard computer-aided detection, classification and tracking system for target identification and a secure digital autonomous communication harness along with AI vision.

SDEL is also developing another AI-enabled autonomous USV, with the key innovation being a multi-level autonomous modular control architecture, which will enable the implementation of navigation algorithms at the desired control level for various missions and weather/ sea conditions. This USV, says Parashar, is especially suitable for missions like special surveillance and reconnaissance operations inside hostile territory, including diving operations and special raids, and counter-terrorism operations.

Both Tardid and SDEL say that the first prototypes of the USVs they are developing are due to be delivered to the navy by August 2023.

Besides USVs, the Indian Navy is also seeking unmanned underwater vehicles—both Remotely Operated Underwater Vehicles (ROUVs) and Autonomous Underwater Vehicles (AUVs). Indigenous UUVs are at an early stage of development, with Larsen & Toubro Defence, Mazagon Shipbuilders Limited (MDL) and DRDO already on the job.

Former eastern naval commander Vice Admiral A.K. Singh says that USVs, UUVs and combat drones are the future for the Indian Navy. "We are behind the US, UK and even Turkish navies in the unmanned vessels domain. The urgent need is for USVs and UUVs to counter mines. The ongoing decline in our submarine force levels is also a cause for concern." If left unaddressed, the navy will be at sea on the high seas. ■